

IET, DAVV, Indore
 B.E. II (IT-A & B)
 Class Test I (Aug'-2023)
 3AIRC1: Discrete Mathematics

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks. Write neatly and properly.

- Q.1. (i) Show that the set Q_1 of all rational numbers other than 1, forms an infinite abelian group (03)
 with the operation '*' defined as: $a * b = a + b - ab$. (02)
- (ii) Define the following with example: Cyclic group, Integral domain.

Q.2. Show that the set of four matrices: (05)

$$M = \left\{ \begin{bmatrix} 1^A & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1^B & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1^C \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0^D & -1 \\ 1 & 0 \end{bmatrix} \right\}, \text{ forms a group with respect to the multiplication}$$

of matrices.

- Q.3. Does the set $F = \{0, 1, 2, 3, 4, 5, 6\}$ forms a field with respect to '+7' and '•7' as the two (05)
 binary compositions? (PTO)